

assignment kine1D - part 1

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:26pm

Quiz Instructions

2 attempt. I keep the last grade. Tutorial in dropbox.



Question 1 4 pts

If the acceleration of an object is negative, the object must be slowing down.

☐

False

☐

True



Question 2 4 pts

Suppose that an object is moving with constant nonzero acceleration. Which of the following is an accurate statement concerning its motion?

☐

In equal times it moves equal distances.

☐

A graph of its velocity as a function of time is a horizontal line.

☐

A graph of its position as a function of time has a constant slope.

☐

In equal times its velocity changes by equal amounts.

☐

In equal times its speed changes by equal amounts.



Question 3 8 pts

As part of an exercise program, a woman walks south at a speed of 2.00 m/s for 60.0 minutes. She then turns around and walks north a distance 3000 m in 25.0 minutes

(a) What is the woman's average velocity during her entire motion?

A) 0.824 m/s south

B) 1.93 m/s south

C) 2.00 m/s south

D) 1.79 m/s south

E) 800 m/s south

(b) What is the woman's average speed during her entire motion?

- A) 0.824 m/s
- B) 1.93 m/s
- C) 2.00 m/s
- D) 1.79 m/s
- E) 800 m/s

1.

☐

(a) B (b) C

☐

(a) D (b) B

☐

(a) A (b) C

☐

(a) E (b) C

⋮

Question 4 4 pts

The position of an object as a function of time is given by $x = bt^2 - ct$, where $b = 2.0 \text{ m/s}^2$ and $c = 6.7 \text{ m/s}$, and x and t are in SI units. What is the instantaneous velocity of the object when $t = 2.2$? hint: the velocity is the derivative of the position at a given time. First take the derivative then plug the time in.

☐

2.7 m/s

☐

1.7 m/s

☐

2.3 m/s

☐

2.1 m/s

⋮

Question 5 4 pts

The position of an object is given by $x = at^3 - bt^2 + ct$ where $a = 4.1 \text{ m/s/s/s}$, $b = 2.2 \text{ m/s/s}$, $c = 1.7 \text{ m/s}$ and x and t are in SI units. What is the instantaneous acceleration of the object when $t = 0.7 \text{ s}$

hint: the acceleration is the derivative of the velocity which is the derivative of the position.

☐

-13 m/s^2

☐

2.9 m/s^2



$$13 \text{ m/s}^2$$



$$4.6 \text{ m/s}^2$$



Question 6 4 pts

The velocity of an object is given by the expression $v(t) = 3.00 \text{ m/s} + (4.00 \text{ m/s}^3)t^2$, where t is in seconds. Determine the position of the object as a function of time if it is located at $x = 1.00 \text{ m}$ at time $t = 0.000 \text{ s}$. hint: take the integral to find the position and there is a constant. The constant is found with the initial condition $t=0 \ x=1$



$$1.00 \text{ m} + (3.00 \text{ m/s})t + (1.33 \text{ m/s}^3)t^3$$



$$1.33 \text{ m}$$



$$(4.00 \text{ m/s})t + 1.00 \text{ m}$$



$$(4.00 \text{ m/s})t$$



$$(3.00 \text{ m/s})t + (1.33 \text{ m/s}^3)t^3$$



Question 7 7.7 pts

Suppose that a car traveling to the west (-x direction) begins to slow down as it approaches a traffic light. Which statement concerning its acceleration must be correct?



Its acceleration is decreasing in magnitude as the car slows down.



Its acceleration is negative.



Its acceleration is zero.



Its acceleration is positive.



Question 8 7.7 pts

What must be your average speed in order to travel 350 km in 5.15 h?



$$67.0 \text{ km/h}$$



$$69.0 \text{ km/h}$$



$$68.0 \text{ km/h}$$



66.0 km/h



Question 9 7.7 pts

An object moves 15.0 m north and then 11.0 m south. Find both the distance it has traveled and the magnitude of its displacement.



26.0 m, 4.0 m



26.0 m, 26.0 m



4.0 m, 26.0 m



4.0 m, 4.0 m



Question 10 7.7 pts

An airplane travels at 300 mi/h south for 2.00 h and then at 250 mi/h north for 750 miles. What is the average speed for the trip?



270 mi/h



275 mi/h



280 mi/h



260 mi/h



Question 11 7.7 pts

A racquetball strikes a wall with a speed of 30 m/s and rebounds in the opposite direction with a speed of 26 m/s. The collision takes 20 ms. What is the average acceleration of the ball during the collision with the wall?

 0 m/s^2  2800 m/s^2  200 m/s^2  1500 m/s^2  1300 m/s^2

**Question 12 7.7 pts**

You are driving home on a weekend from school at 55 mi/h for 110 miles. It then starts to snow and you slow to 35 mi/h. You arrive home after driving 4 hours and 15 minutes. How far is your hometown from school?

☐

180 mi

☐

210 mi

☐

190 mi

☐

200 mi

**Question 13 7.7 pts**

A polar bear starts at the North Pole. It travels 1.0 km south, then 1.0 km east, and then 1.0 km north to return to its starting point. This trip takes 45 min. What was the bear's average velocity?

☐

4.0 km/h

☐

0.00 km/h

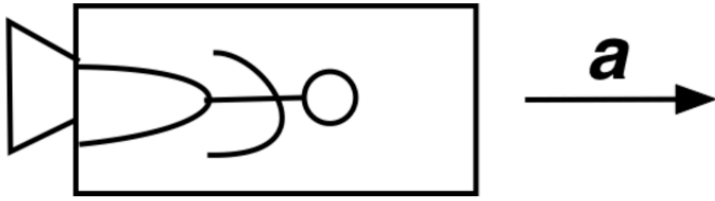
☐

5.3 km/h

☐

0.067 km/h

**Question 14 4 pts**



Science fiction writers imagine a ship in deep space, accelerating at a constant 9.8 m.s^{-2} , which gives the travellers a sensation similar to the weight they feel on the Earth, where falling objects accelerate at $g = 9.8 \text{ m.s}^{-2}$. Starting from rest (and ignoring effects due to special relativity), how long does it take such a ship to accelerate to 0.1 times the speed of light c (recalling that $c = 3.0 \times 10^8 \text{ m.s}^{-1}$), and how far does it travel during that time? (Note: 0.1 has only 1 sig fig!)

Answer: The ship travels for _____ and travels a distance of _____ .

Round to get 1 sig fig. The distance is in km (1 km=1000m). the time is in days (1 day=24x 3600s)

☐

4 days, $5 \times 10^{14} \text{ km}$

☐

4 days, $5 \times 10^8 \text{ km}$

☐

40 days, $5 \times 10^{10} \text{ km}$

☐

40 days, $5 \times 10^4 \text{ km}$

☐

Question 15 6 pts

A value commonly taken for the perception-reaction time of motorists is $\Delta t = 1.5 \text{ s}$. Using this value, determine the total stopping distance for a car travelling at $30 \text{ m} \cdot \text{s}^{-1}$ (two significant figures). In calculating the stopping distance, let's assume that the vehicle continues at its initial speed for Δt , then decelerates to rest. A reasonable value for the deceleration in braking is (constant) $8.0 \text{ m} \cdot \text{s}^{-2}$.

Stopping distance = _____ metres. (Hint: How many significant figures?)

hint: its a bit tricky. Draw the situation first and understand the wording. Round to get 1 sig fig. Think !

☐

about 60m

☐

about 20m

☐

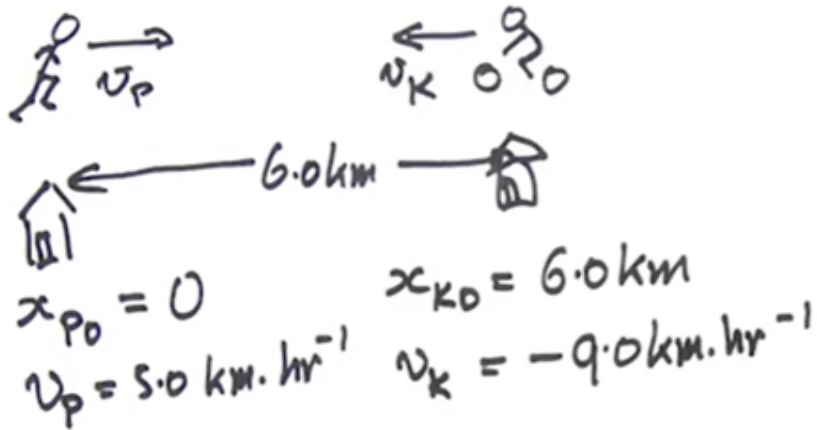
about 100m

☐

about 50m

☐

Question 16 6 pts



Patrick runs. Kevin bikes. They move toward each other and their houses are 6km apart. How long will it take for them to meet? about?



about 10s



about 1 hour



about 40min



about 25min



Question 17 4 pts

You sing a suitably chosen tune at 84 beats per minute and walk (in time, exactly one step per beat) with a step length of 0.90 metres. What is your speed in km/hr? (Hint: use the multiply by one technique from week 1. It's worthwhile writing this out fully.)



Your speed is _____ km/hr. (Hint: how many significant figures?)

hint: first you can find your speed in m/min then convert to km/h.

Remember conversions?

$$1 \text{ m/min} = (1 \text{ m} / \text{min}) (\text{km}/1000) (60 \text{ min}/1 \text{ hour})$$



4.5 km/h



20km/h



1 km/h



80 km/h

Not saved

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